

Foundational Geometry:

Invariants Across Formalisms and Structures Preceding a Geometric Description

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Abstract

We ask which claims about “structures that precede a geometric description” can be formulated and tested inside explicitly defined mathematical frameworks. We introduce a minimal bookkeeping language: a *formalism* is a class of structures, a *translation* is a map between formalisms, and an *invariant across formalisms* is a quantity that factors through a translation. Geometry then appears as what survives the passage from a richer to a poorer formalism, in the spirit of Klein’s Erlangen programme and of the meaningfulness criterion of measurement theory. The language is worked out in full on one chain where every step can be proved or computed, the finite Euclidean chain $\text{coordinates} \rightarrow \text{distance matrix} \rightarrow \text{ranking of distances} \rightarrow \text{betweenness and congruence}$. Three results are established. (i) On generic finite samples the relational primitives of elementary geometry are empty: exact integer arithmetic over 4000 random configurations produced no non-degenerate instance of betweenness or congruence, while integer grids produce dozens to thousands; the primitives that axiomatise the continuum are therefore not sample-robust. (ii) The ordinal formalism is strictly coarser than similarity geometry for every finite $n \geq 3$, yet a non-metric reconstruction from the ranking alone recovers the configuration up to similarity with a median Procrustes disparity of 2.1×10^{-10} at $n = 256$ in the plane, 7.6 orders of magnitude below its value at $n = 8$; this is the finite-sample face of the asymptotic uniqueness theorem for ordinal embeddings, and the class itself is measured directly: its inradius is one quarter of the minimal gap between consecutive sorted distances, a quantity that scales empirically as $n^{-4.2}$. (iii) Standard geometric quantities sort themselves by *formalism depth*: minimum spanning tree, nearest-neighbour graph, relative neighbourhood graph and diameter pair are ordinal invariants, whereas diameter, affine dimension and the Gabriel graph are metric but not ordinal invariants at finite n . The same language is then applied to the causal-order translation $\text{Lorentzian coordinates} \rightarrow \text{causal order}$, followed by a statistical step that adds counting, on uniform sprinklings of the 1+1 Minkowski diamond, where the kernel of the order translation contains the conformal group, is strictly larger at finite n , and counting restores proper times and coordinates. The paper claims no unified theory and reports no experimental physics; its contribution is to turn a recurring foundational phrase into a list of statements with explicit status.

1 Introduction

The idea that geometry is not primitive, that something poorer than a metric or a coordinate system comes first and geometry is recovered from it, is old and recurs in several disciplines under different names. Klein’s Erlangen programme defines a geometry by the group of transformations under which its statements are invariant [12]. Hilbert and Tarski axiomatise Euclidean geometry from the primitive relations of betweenness and congruence [11, 30, 31]. Menger and Blumenthal ask which metric spaces are “Euclidean” and how much of a configuration a distance table determines [2, 20]. Measurement theory calls a statement *meaningful* when it is invariant under the admissible transformations of the scale on which the quantities were measured [14, 23, 29]. In

relativity, the causal order of a spacetime determines its topology and its conformal geometry [10, 19], and the causal set programme [5] condenses this, in Sorkin’s later slogan, into “order plus number equals geometry” [28].

Each of these traditions is precise on its own ground, but the phrase “structure preceding geometry” is often used across them without saying which structure, which geometry, and in what sense one precedes the other. The research line to which this draft belongs (MRT001) had, at the time of writing, the modest and explicit objective of *specifying which claims of this kind can be formulated and tested within defined mathematical frameworks*. This paper is an attempt to discharge that objective by three moves.

1. We introduce a small bookkeeping language (Section 2): formalisms, translations between them, the kernel of a translation, invariants along a translation, and the *formalism depth* of a quantity. Nothing in it is new; its purpose is to make “ G precedes F ” a predicate that can be checked.
2. We work the language out completely on one chain where every claim can be proved or computed: finite configurations of points in $\mathbb{R}^d$, read successively as coordinates, as a distance matrix, as a ranking of the pairwise distances, and as Tarski’s betweenness and congruence relations (Sections 3 and 4).
3. We apply the same language to the causal order of a uniform sprinkling of 1+1 Minkowski space (Section 5), where the continuum theorems of Hawking–King–McCarthy and Malament have finite, computable counterparts.

Scope. We do not propose a unified theory, a new geometry or a physical model; the Lorentzian part is a mathematical model with a fixed background and is used only because it is the sharpest known instance of “order before metric”. Every numerical statement in the paper is produced by two short scripts included with the manuscript, with a fixed seed (Appendix A). Section 6 lists all claims with their status: proved here, classical, taken from the literature, computationally verified, or conjectural.

2 Formalisms, translations and invariants

Definition 2.1 (Formalism). A *formalism* is a pair $F = (\text{Obj}_F, \cong_F)$ consisting of a class of objects and an equivalence relation on it, read as isomorphism. Two objects that are isomorphic are regarded as the same description.

Definition 2.2 (Translation, kernel). A *translation* $\tau: F \rightarrow G$ is a map $\text{Obj}_F \rightarrow \text{Obj}_G$ such that $X \cong_F Y$ implies $\tau X \cong_G \tau Y$. Its *kernel* is the equivalence relation $X \sim_\tau Y \iff \tau X \cong_G \tau Y$ on Obj_F , which is coarser than (contains) $\cong_F$. Two objects in the same kernel class are *indistinguishable in G* .

Definition 2.3 (Invariant across formalisms). A function Φ on Obj_F is an *invariant along τ* , or a G -invariant of F , if $X \sim_\tau Y$ implies $\Phi(X) = \Phi(Y)$; equivalently, Φ factors through τ . We write $\text{Inv}(\tau)$ for the set of such functions.

Definition 2.4 (Chain, formalism depth). A *chain* is a sequence of translations $F_0 \xrightarrow{\tau_1} F_1 \xrightarrow{\tau_2} \dots \xrightarrow{\tau_k} F_k$. The *formalism depth* of a quantity Φ on Obj_{F_0} is the largest i such that $\Phi \in \text{Inv}(\tau_i \circ \dots \circ \tau_1)$, that is, the poorest formalism in the chain from which Φ can still be computed; the empty composition ($i = 0$) is allowed, so a quantity that is not invariant even along τ_1 has depth 0, the coordinate level.

Lemma 2.5 (Monotonicity). *For composable translations, $\text{Inv}(\tau_2 \circ \tau_1) \subseteq \text{Inv}(\tau_1)$. Hence along a chain the sets of invariants decrease and the kernels increase.*

Proof. If $\tau_1 X \cong \tau_1 Y$ then $\tau_2 \tau_1 X \cong \tau_2 \tau_1 Y$, so $\sim_{\tau_1} \subseteq \sim_{\tau_2 \tau_1}$, and a function constant on the classes of the coarser relation is constant on those of the finer one. $\square$

Definition 2.6 (Faithfulness, precedence). Let $\tau: F \rightarrow G$ and let $\mathcal{C} \subseteq \text{Obj}_F$ be a class of objects of interest. The translation is *faithful on $\mathcal{C}$ up to a group H* acting on Obj_F if, for $X, Y \in \mathcal{C}$, $X \sim_\tau Y$ implies $Y = hX$ for some $h \in H$. For a family of formalisms F_n (below, $\text{Coord}(n, d)$), classes $\mathcal{C}_n \subseteq \text{Obj}_{F_n}$ and a fixed H -invariant distance δ on Obj_{F_n} (below, the Procrustes disparity), translations τ_n are *asymptotically faithful up to H* on $(\mathcal{C}_n)$ if $\sup\{\delta(X, Y) : X \in \mathcal{C}_n, Y \sim_{\tau_n} X\} \rightarrow 0$ as $n \rightarrow \infty$. We say that G *precedes F* on $\mathcal{C}$ when τ has a non-trivial kernel and is nevertheless faithful, or asymptotically faithful, up to a group H that is itself a symmetry group of F .

The definition makes the foundational phrase testable: to claim that a structure G precedes a geometry F one must exhibit the translation, compute or bound its kernel, and prove or measure its faithfulness. Two classical readings are recovered at once.

Remark 2.7 (Erlangen). If Obj_F is a set with an action of a group H and τ is the quotient map to Obj_F/H , then $\text{Inv}(\tau)$ is exactly the set of H -invariants, Klein's geometry of the pair (Obj_F, H) . Euclidean, similarity, affine and projective geometry are the invariants of successively coarser quotients, and Lemma 2.5 is the Erlangen hierarchy.

Remark 2.8 (Meaningfulness). Stevens' scale types are kernels of translations [29]. A ratio scale identifies quantities up to a positive factor, an interval scale up to a positive affine map, an ordinal scale up to a monotone bijection. A statement is *meaningful* in the sense of Krantz et al. [14] and Narens [23] precisely when it is an invariant along the corresponding translation. The chain studied below realises two of these scale types on one object: the distance matrix is a ratio-scale description of a configuration and the ranking of distances is its ordinal-scale description; a configuration read up to similarity is what the ratio scale determines.

3 The finite Euclidean chain

Fix $n \geq 3$ and $d \geq 1$. Write $[n] = \{1, \dots, n\}$ and P_n for the set of unordered pairs of distinct elements of $[n]$, $m = |P_n| = \binom{n}{2}$. All objects are labelled; quotienting by relabelling adds a graph-isomorphism layer that is orthogonal to the questions asked here.

- $\text{Coord}(n, d)$: injective configurations $X = (x_1, \dots, x_n) \in (\mathbb{R}^d)^n$ (distinct points), with equality as isomorphism.
- $\text{Met}(n)$: symmetric matrices $D \in \mathbb{R}^{n \times n}$ with zero diagonal and positive off-diagonal entries, with equality as isomorphism.
- $\text{Ord}(n)$: total preorders $\preceq$ on P_n (generically total orders), with equality as isomorphism.
- $\text{Rel}(n)$: pairs (B, C) of a ternary relation $B \subseteq [n]^3$ (betweenness) and a binary relation $C \subseteq P_n \times P_n$ (congruence), with equality as isomorphism.

The translations are the obvious ones. $\tau_1: \text{Coord} \rightarrow \text{Met}$ sends X to its distance matrix $D_X = (\|x_a - x_b\|)_{a,b}$. $\tau_2: \text{Met} \rightarrow \text{Ord}$ sends D to the preorder $\{a, b\} \preceq \{c, d\} \iff D_{ab} \leq D_{cd}$. $\tau_3: \text{Met} \rightarrow \text{Rel}$ sends D to

$$B_D = \{(a, b, c) \text{ distinct} : D_{ab} + D_{bc} = D_{ac}\}, \quad C_D = \{(p, q) \in P_n^2 : p \neq q, D_p = D_q\}.$$

The congruence part of τ_3 factors through τ_2 (congruent pairs are ties of the preorder) but the betweenness part does not, since it involves an additive equality; the chain therefore branches at Met .

3.1 Coordinates to metric: the classical step

Theorem 3.1 (Schoenberg; Young and Householder). *Two labelled configurations $X, Y \in (\mathbb{R}^d)^n$ have the same distance matrix if and only if $Y = gX$ for a Euclidean motion $g \in E(d)$, possibly improper. A matrix D is the distance matrix of a configuration in $\mathbb{R}^d$ if and only if $-\frac{1}{2}JD^{(2)}J$ is positive semidefinite of rank at most d , where $D^{(2)}$ is the entrywise square and $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.*

See Blumenthal [2], Schoenberg [25], Young and Householder [36]. In the language of Section 2: $\ker \tau_1$ is the orbit relation of $E(d)$, τ_1 is faithful up to $E(d)$ on all of Coord , and $\text{Inv}(\tau_1)$ is Euclidean geometry in Klein’s sense. The metric formalism precedes the coordinate one in the strongest possible way: nothing is lost except position and orientation.

3.2 Metric to relations: the primitives are empty on samples

Tarski’s system axiomatises elementary Euclidean geometry with the single sort “point” and the two primitives betweenness and congruence, and the system is complete and decidable [30, 31]. The restriction of these primitives to a finite sample behaves very differently from their behaviour on the continuum.

Proposition 3.2 (Generic emptiness). *Let $d \geq 2$ and $n \geq 3$, and let $\mathcal{G} \subseteq (\mathbb{R}^d)^n$ be the set of configurations in which no three points are collinear and all m pairwise distances are distinct. Then $\mathcal{G}$ is open, dense and of full Lebesgue measure, and for every $X \in \mathcal{G}$*

$$B_{D_X} = \emptyset \quad \text{and} \quad C_{D_X} = \emptyset.$$

Consequently $\tau_3 \circ \tau_1$ is constant on $\mathcal{G}$: its kernel identifies all generic configurations of a given size, and every relational invariant is constant on $\mathcal{G}$.

Proof. Collinearity of three distinct points x_a, x_b, x_c is the vanishing of all 2×2 minors of the $d \times 2$ matrix $[x_b - x_a \mid x_c - x_a]$, a system of polynomial equations in the coordinates that is not identically satisfied when $d \geq 2$; its solution set is a proper real-algebraic subset, hence closed, nowhere dense and null. Equality $\|x_a - x_b\|^2 = \|x_c - x_d\|^2$ for two distinct pairs is a single non-constant polynomial equation (non-constant because moving x_a alone changes the left side and, when $a \notin \{c, d\}$, not the right; when the pairs share a point the same argument applies to the point that is not shared), with the same consequences. $\mathcal{G}$ is the complement of a finite union of such sets, so it is open, dense and of full measure. For $X \in \mathcal{G}$: if $D_{ab} + D_{bc} = D_{ac}$ with a, b, c distinct then, by the equality case of the triangle inequality for the Euclidean norm (strict convexity), $x_b - x_a$ and $x_c - x_b$ are non-negative multiples of one another, so x_b lies on the segment $[x_a, x_c]$ and the three points are collinear, contradicting $X \in \mathcal{G}$. A congruence (p, q) with $p \neq q$ contradicts distinctness of the distances. The final assertion follows because $\tau_3 \tau_1 X = (\emptyset, \emptyset)$ for all $X \in \mathcal{G}$. $\square$

Remark 3.3 (The line). For $d = 1$ every triple is collinear and B_{D_X} is the betweenness relation of the linear order of the sample, which is far from empty; the proposition then holds only for C . On the line the relational formalism carries exactly the order type of the sample, which is the structure Section 5 takes as its starting point.

Computational check (E1). Proposition 3.2 is a statement about a full-measure set, and a floating-point test of the equalities defining B and C would be misleading: the betweenness gap $D_{ab} + D_{bc} - D_{ac}$ is quadratic in the distance from x_b to the line through x_a, x_c , so a relative tolerance of 10^{-9} admits every triple whose middle point lies within about $10^{-4.5}$ of a line and, at $n = 50$, produced of the order of one spurious triple per configuration in a preliminary floating-point run. We therefore sample integer coordinates uniformly in $\{0, \dots, 2^{30} - 1\}^d$ and test collinearity and equality of squared distances *exactly* in 64-bit integer arithmetic (the

n	d	configurations	betweenness instances	congruence instances
5	2	500	0	0
10	2	500	0	0
20	2	500	0	0
50	2	500	0	0
5	3	500	0	0
10	3	500	0	0
20	3	500	0	0
50	3	500	0	0
grid		n	betweenness instances	congruence instances
3×3	9	8	138	
4×4	16	44	976	
5×5	25	152	4242	

Table 1: E1. Exact counts of non-degenerate betweenness triples and congruent pair-pairs in random integer configurations (upper panel, 500 configurations per row) and in square integer grids (lower panel). A betweenness instance is a pair $(b, \{a, c\})$ with b strictly between a and c (a row of four collinear points contributes $1 + 1 + 2 = 4$); a congruence instance is an unordered pair of distinct point-pairs at equal distance.

sampling grid is fine enough that the result is the generic one with overwhelming probability, and coarse enough that an exact coincidence, should one occur, would be counted). Table 1 reports the totals: across 8 settings and 4000 configurations, 0 betweenness instances and 0 congruence instances were found. The integer grids in the lower panel, where the relations are rich, show that the test itself is not blind.

Remark 3.4 (Axiomatic versus observational primitives). The primitives that make Tarski’s theory complete on the plane carry no information about a generic finite sample of the plane. Any empirical test of a geometric claim uses finitely many measured points, so such a test must pass through the metric or the ordinal formalism; betweenness and congruence can only be *inferred* from them, with a tolerance whose meaning depends on n and on the chosen tolerance, not on the configuration. This is the first precise sense in which “the structure that precedes geometry” cannot be the axiomatic primitives.

3.3 Metric to ordinal: finite non-uniqueness

Proposition 3.5 (Finite non-uniqueness). *Let $n \geq 3$, $d \geq 1$. For every configuration X of distinct points whose pairwise distances are distinct, the kernel class of X under $\tau_2 \circ \tau_1$ contains a neighbourhood of X in $(\mathbb{R}^d)^n$, and that neighbourhood contains configurations that are not similar to X . In particular $\tau_2 \tau_1$ is not faithful up to $\text{Sim}(d)$ on any open set, for any finite n .*

Proof. The distances are continuous functions of X , and the ordinal pattern of X is defined by finitely many strict inequalities between them, which persist in a neighbourhood U of X ; so U lies in the kernel class. Pick three labels a, b, c and move x_b along the ray from x_a through x_b by a small amount: D_{ab} changes while D_{ac} does not, so the ratio D_{ab}/D_{ac} changes. Similarities preserve all distance ratios, hence the moved configuration, which lies in U for a small enough move, is not similar to X . $\square$

Example 3.6 (E3). On the line, $\{0, 1, 3\}$ and $\{0, 1, 2.5\}$ have the same ordinal pattern (distances $1 < 2 < 3$ and $1 < 1.5 < 2.5$ in the same pair order) and are not similar. In the plane, the triangles with sides $(3, 4, 5)$ and $(3, 4, 6)$ have the same pattern and angles $(36.9, 53.1, 90.0)$ and $(26.4, 36.3, 117.3)$ degrees respectively. For $n = 3$ the ordinal formalism records exactly the order of the three sides, equivalently the order of the three angles, and its generic fibres are the six open cells into which these orderings cut the two-dimensional shape space of triangles.

The ordinal formalism nevertheless retains dimensional information. For $n = 4$ there are 720 possible orderings of the six distances. Sampling 300 000 uniform configurations, we observe 120 distinct orderings on the line but all 720 in the plane and all 720 in space (the least frequent planar ordering was seen 53 times, so the count is not an artefact of sample size). The count on the line has a short explanation: for sorted points with gaps u, v, w the six distances are $u, v, w, u + v, v + w, u + v + w$, and only ten orderings of these are consistent with positivity, each labelled pattern arising from two (labelling, gap-order) combinations under reflection, giving $24 \cdot 10/2 = 120$.

3.4 Metric to ordinal: asymptotic faithfulness

Theorem 3.7 (Kleindessner and von Luxburg; informal statement). *Let $\Omega \subset \mathbb{R}^d$ be bounded, connected and open, and let $x_1, x_2, \dots$ be points that become dense in Ω . Then the ordinal pattern of $(x_1, \dots, x_n)$ determines the configuration up to similarity in the limit: any sequence of configurations $y^{(n)} \in (\mathbb{R}^d)^n$ with the same ordinal pattern as $(x_1, \dots, x_n)$ converges, after alignment by similarity transformations, to the original points as $n \rightarrow \infty$, in the sense made precise in Kleindessner and von Luxburg [13].*

Terada and von Luxburg [32] show that the same conclusion holds when only the unweighted k -nearest-neighbour graph is retained, for points drawn from a density and $k \rightarrow \infty$ with $k/n \rightarrow 0$. In our language: $\tau_2\tau_1$ is asymptotically faithful up to $\text{Sim}(d)$ on dense samples of bounded domains, so the ordinal formalism precedes similarity geometry, although by Proposition 3.5 it never does so exactly at finite n .

Computational check (E2). For $d \in \{2, 3\}$ and n from 8 to 256 (128 for $d = 3$) we sample n points uniformly in the unit cube, compute the rank matrix of the pairwise distances (so that *only* the ordinal pattern enters), reconstruct a configuration by non-metric multidimensional scaling [17, 27] as implemented in scikit-learn [24] (SMACOF with isotonic regression, four random starts, stress tolerance 10^{-13} , at most 50 000 iterations), and measure the Procrustes disparity between the original and the reconstruction after optimal translation, rotation, reflection and scaling [9, 26]: both configurations are centred and scaled to unit Frobenius norm and the disparity is the residual sum of squares, a similarity-invariant number in $[0, 1]$. Table 2 and Figure 1 report 255 runs. In the plane the median disparity falls from 8.2×10^{-3} at $n = 8$ to 2.1×10^{-10} at $n = 256$, 7.6 orders of magnitude; in space from 4.2×10^{-2} to 4.1×10^{-9} at $n = 128$. The local exponents between successive doublings of n (last column of Table 2) range from 6.7 down to 3.0 in the plane, so a single power law is not supported. Two cautions apply. First, the reconstruction is only approximately in the kernel class: the solver stops at a stress tolerance, and the Kendall discordance between the input ranking and the ranking of the output, that is, the fraction of ordinal constraints the output violates, is 4.3×10^{-6} at $n = 256$ (Table 2); with the looser tolerance 10^{-9} of a preliminary run the disparities were between ten and more than a hundred times larger, depending on n , so the absolute values are partly set by the stopping rule. Second, since X itself lies in the class, a solver output near X measures the distance from X to one approximate member of the class, not the class’s diameter. The numbers are consistent with a rapidly shrinking class, because the solver starts from a random configuration and sees only the ranking, but they do not measure the class; they are consistent with Theorem 3.7, not its quantitative content.

Invariance check (E2b). A monotone distortion $D \mapsto D^p$, $p \in [2, 4]$, leaves the rank matrix literally unchanged, so non-metric scaling, which reads only the rank matrix, returns the identical reconstruction by construction (the reported maximum disparity between the two outputs, 1.9×10^{-31} , is round-off of the alignment); this part of E2b checks the pipeline, not a fact about geometry. The content is in the metric column: metric multidimensional scaling,

d	n	runs	median	Q_1	Q_3	max	discordance	local exp.
2	8	30	8.2×10^{-3}	5.8×10^{-3}	1.6×10^{-2}	1.6×10^{-1}	2.1×10^{-2}	–
2	16	30	8.1×10^{-5}	5.4×10^{-5}	1.2×10^{-4}	3.0×10^{-4}	2.4×10^{-3}	6.7
2	32	30	1.8×10^{-6}	1.3×10^{-6}	2.2×10^{-6}	3.3×10^{-6}	3.1×10^{-4}	5.5
2	64	30	3.0×10^{-8}	2.7×10^{-8}	3.6×10^{-8}	4.9×10^{-8}	5.1×10^{-5}	5.9
2	128	30	1.6×10^{-9}	1.5×10^{-9}	1.7×10^{-9}	2.0×10^{-9}	1.2×10^{-5}	4.3
2	256	30	2.1×10^{-10}	2.0×10^{-10}	2.2×10^{-10}	2.3×10^{-10}	4.3×10^{-6}	3.0
3	8	15	4.2×10^{-2}	1.6×10^{-2}	7.5×10^{-2}	1.9×10^{-1}	3.7×10^{-2}	–
3	16	15	3.8×10^{-4}	2.8×10^{-4}	4.9×10^{-4}	6.6×10^{-4}	3.9×10^{-3}	6.8
3	32	15	4.2×10^{-6}	3.3×10^{-6}	5.8×10^{-6}	8.3×10^{-6}	4.6×10^{-4}	6.5
3	64	15	8.3×10^{-8}	7.9×10^{-8}	9.0×10^{-8}	1.2×10^{-7}	7.0×10^{-5}	5.7
3	128	15	4.1×10^{-9}	3.8×10^{-9}	4.4×10^{-9}	4.7×10^{-9}	1.7×10^{-5}	4.3

Table 2: E2. Procrustes disparity (similarity-invariant, in $[0, 1]$) between a uniform random configuration and its reconstruction from the ranking of the pairwise distances alone. Discordance: median fraction of ordinal constraints violated by the reconstruction (Kendall). Local exp.: $-\log(\text{median}_n / \text{median}_{n/2}) / \log 2$.

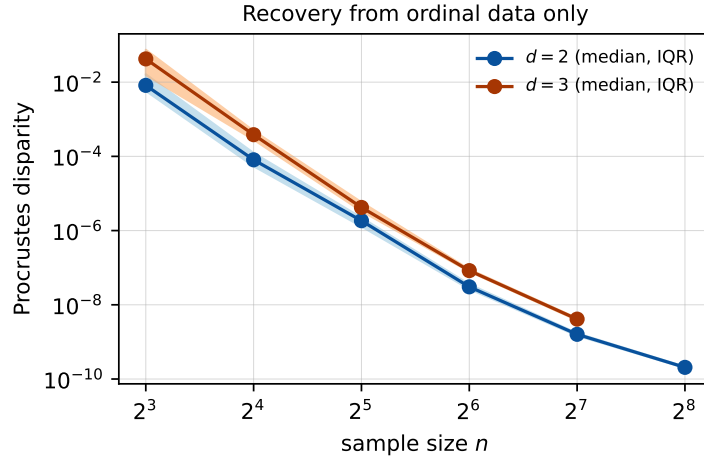


Figure 1: E2. Median and interquartile range of the Procrustes disparity of the ordinal reconstruction as a function of n , for $d = 2$ and $d = 3$.

which uses the values, reconstructs X from D with disparity 7.9×10^{-13} but from D^p with median disparity 4.5×10^{-2} (minimum 1.3×10^{-2} ; $n = 64$, 20 trials). The distortion changes the object of Met and not the object of Ord; this is what “invariant across formalisms” means operationally.

3.5 Measuring the ordinal class directly

The cautions attached to E2 can be met by measuring the kernel class itself rather than a solver’s approximation to it. One measurement is exact.

Proposition 3.8 (Thickness of the ordinal class). *Let $X \in \text{Coord}(n, d)$ have distinct pairwise distances and let $g(X) > 0$ be the minimal difference between consecutive values in the sorted list of its m pairwise distances. (i) If $\|y_i - x_i\| < g(X)/4$ for all i , then Y has the same ordinal pattern as X . (ii) If the two pairs of points realising $g(X)$ are disjoint, there is a configuration Y with $\max_i \|y_i - x_i\| = g(X)/4$ in which those two distances coincide, so that every neighbourhood of Y contains configurations with a different pattern. Consequently the inradius of the kernel class at X , in the norm $\max_i \|y_i - x_i\|$, lies in $[g(X)/4, g(X)/2]$, and equals $g(X)/4$ in case (ii).*

Proof. (i) By the triangle inequality, $|\|y_a - y_b\| - \|x_a - x_b\|| \leq \|y_a - x_a\| + \|y_b - x_b\| < g/2$ for every pair, so every distance moves by less than $g/2$ and the difference between any two

n	configs	median g	disjoint	kept $< g/4$	changed at $g/4$	median walk radius
8	20	6.8×10^{-4}	9/20	400/400	9/9	5.6×10^{-3}
16	20	3.5×10^{-5}	17/20	400/400	17/17	4.8×10^{-5}
32	20	1.6×10^{-6}	17/20	400/400	17/17	2.6×10^{-7}
64	20	2.4×10^{-7}	17/20	400/400	17/17	1.0×10^{-9}
128	20	6.3×10^{-9}	20/20	400/400	20/20	3.6×10^{-12}
256	12	2.9×10^{-10}	12/12	240/240	11/12	5.5×10^{-15}

Table 3: E2c. Median minimal gap g between consecutive sorted distances (the inradius of the class is $g/4$ when the extremal pairs are disjoint), the numerical checks of Proposition 3.8, and the median of the farthest class member found by a constrained random walk (Procrustes disparity from X , a lower bound on the class radius). Walk lengths decrease from 8000 steps at $n = 8$ to 1500 at $n = 256$; interquartile ranges are shown in Figure 2 and listed in the result files.

distances changes by less than g ; two distances that are consecutive in the sorted order differ by at least g and therefore keep their order, hence so do all pairs. (ii) Let $\{a, b\}$ and $\{c, d\}$ be the disjoint pairs with $D_{cd} - D_{ab} = g$. Move x_a and x_b apart along the line through them by $g/4$ each, and x_c and x_d towards each other along their line by $g/4$ each, leaving the other points fixed: the first distance becomes $D_{ab} + g/2$ and the second $D_{cd} - g/2 = D_{ab} + g/2$, a tie, and the four displacements have norm $g/4$. For the upper bound in general, two distinct pairs share at most one point; if $\{a, b\}$ and $\{a, c\}$ realise g , moving x_b away from x_a and x_c towards x_a by $g/2$ each produces the same tie. $\square$

Computational check (E2c). For $d = 2$ and n from 8 to 256, 112 uniform configurations were examined (Table 3, Figure 2). Random perturbations of every point by less than $g/4$ preserved the pattern in all 2240 of 2240 trials, and the explicit displacement of part (ii) changed it in all 91 of 92 configurations whose extremal pairs were disjoint. The median gap scales as $n^{-4.2}$, which is what one expects for the minimal spacing among $m \approx n^2/2$ weakly dependent values with a bounded density, of order $1/m^2 \propto n^{-4}$: the class is thin because n points carry $n^2/2$ distances. A second, similarity-invariant measurement is a random walk constrained to the class (proposals accepted only if the ranking is unchanged), which records the farthest member found in Procrustes disparity; this is a lower bound on the radius of the class at X in that metric. Its median scales as $n^{-8.0}$, consistent with the disparity being quadratic in the displacement, and at $n = 64$ it is 1.0×10^{-9} , two orders of magnitude *below* the disparity 3.0×10^{-8} at which the E2 solver stopped with discordance 5.1×10^{-5} . The solver’s residual therefore reflects approximate membership, not the size of the class; the class is smaller than E2 alone could show, and the two measurements together give the finite- n content of Theorem 3.7 for this sampling distribution as lower bounds on how fast the class shrinks.

4 Formalism depth of geometric quantities

Lemma 2.5 orders the invariants of the chain, and Sections 3 locate the boundaries: position is lost at τ_1 , scale at τ_2 , everything at τ_3 on generic samples. Table 4 classifies familiar quantities by the poorest formalism from which they can be computed, together with a direct computational check (E4): each quantity is computed for 300 random configurations of 30 points in the plane, then recomputed after a random rigid motion, after a random similarity, and after a random monotone distortion $D \mapsto D^p$, $p \in [0.3, 3]$, and the fraction of trials in which the value is preserved is reported.

The ordinal rows have one-line proofs. The diameter pair is the argument of a maximum, which depends only on the order. The k -nearest-neighbour graph is defined by comparisons $D_{ij} < D_{ik}$ within each row. The minimum spanning tree is produced by Kruskal’s algorithm from the

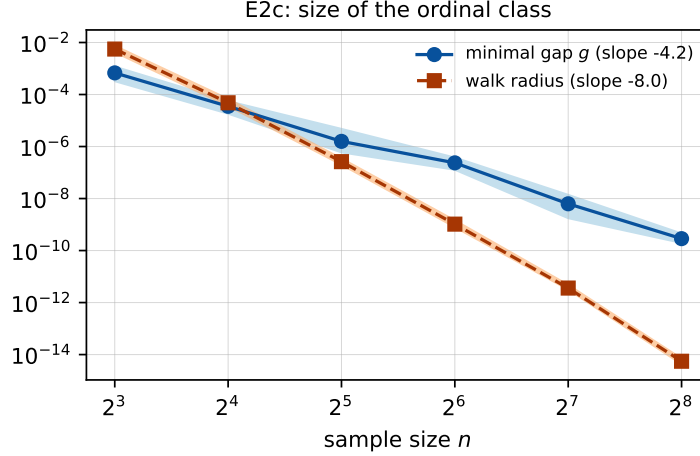


Figure 2: E2c. Median and interquartile range of the minimal gap g and of the walk radius against n ($d = 2$).

quantity	depth	rigid motion	similarity	monotone distortion
centroid (position)	coordinate	0.00	0.00	–
diameter (value)	metric	1.00	0.00	0.00
affine dimension	metric	1.00	1.00	0.35
Gabriel graph	metric	1.00	1.00	0.03
diameter pair	ordinal	1.00	1.00	1.00
k -nearest-neighbour graph ($k = 3$)	ordinal	1.00	1.00	1.00
minimum spanning tree	ordinal	1.00	1.00	1.00
relative neighbourhood graph	ordinal	1.00	1.00	1.00

Table 4: E4. Formalism depth of standard quantities and the fraction of 300 trials in which each is preserved under rigid motions (which generate $\ker \tau_1$), similarities, and monotone distortions of the distance matrix (which generate $\ker \tau_2$ on **Met**; the distorted matrix is in general not Euclidean, so this column tests the defining formula on **Met**, a sufficient condition for ordinal depth). The centroid is undefined for a bare metric.

sorted list of edge weights and is therefore invariant under any strictly monotone reweighting [16]. The relative neighbourhood graph [33] contains ij iff no k satisfies $\max(D_{ik}, D_{jk}) < D_{ij}$, again a comparison. The metric rows fail for equally simple reasons: the diameter changes under scaling; the affine dimension is the number of positive eigenvalues of the doubly centred squared-distance matrix (Theorem 3.1), and a power D^p of a planar metric is not planar: for $p < 1$ it is a Euclidean metric in dimension $n - 1$ (Schoenberg), for $p > 2$ the centred matrix has many positive eigenvalues, and only for $1 < p < 2$ did it have exactly two positive eigenvalues in every trial, so the value is preserved in a fraction 0.35 of trials, essentially the probability that $p \in (1, 2)$; the Gabriel graph [8] contains ij iff no k satisfies $D_{ik}^2 + D_{jk}^2 < D_{ij}^2$, an additive condition, and is preserved in a fraction 0.03 of trials. That the relative neighbourhood graph is ordinal and the Gabriel graph is not, although both are “proximity graphs” of the same family, illustrates that depth is a property of the defining formula, not of the intuitive kind of the quantity. The metric rows are also genuinely non-ordinal on **Coord**, which the distortion column alone does not show, since the distorted matrix leaves the Euclidean locus: a collinear configuration with distinct distances and a small generic perturbation of it share the ordinal pattern (Proposition 3.5) and have affine dimensions 1 and 2; a triangle with a right angle at k , perturbed to acute or to obtuse without changing the order of its sides, keeps the pattern and gains or loses the Gabriel edge ij . Both witnesses are checked in the code (E4b: patterns equal, yes, dimensions 1 and 2; patterns equal, yes, edge present yes and no).

Theorem 3.7 adds an asymptotic column to the table: every *continuous* similarity invariant (distance ratios, angles) is asymptotically determined by the ordinal pattern, because the pattern determines the configuration up to similarity in the limit; for discrete invariants such as the Gabriel graph the statement holds only away from configurations near the defining equality, since a dense sample always contains near-borderline pairs. The finite- n classification and the asymptotic one therefore differ, and a claim that “order determines geometry” must say which of the two is meant.

5 A second translation: causal order and counting

The Euclidean chain has an analogue in which the poorer structure is a partial order rather than a ranking. Consider the causal diamond of 1+1 Minkowski space in null coordinates $u = t - x$, $v = t + x$, so that $ds^2 = du dv$, the diamond is the unit square $(0, 1)^2$, and the proper time between causally related events is $\tau(p, q) = \sqrt{(u_q - u_p)(v_q - v_p)}$. The causal order is the product order, $p \prec q \iff u_p < u_q$ and $v_p < v_q$; unrelated pairs are written $p \parallel q$. The formalisms are:

- $\text{Coord}_{1,1}(n)$: labelled configurations $(u_i, v_i)_{i \leq n}$ in the diamond;
- $\text{Caus}(n)$: partial orders on $[n]$ (labelled); the image of σ_1 below is the set of labelled orders of Dushnik–Miller dimension at most two [7], the orders realised by two linear orders, here $<_u$ and $<_v$;
- $\text{Caus+N}(n)$: the same partial order together with the statistical hypothesis that the configuration is a uniform sample of the diamond, so that counts of points estimate areas; its “invariants” are estimators, consistent as $n \rightarrow \infty$, rather than functions factoring through a translation.

The translation $\sigma_1: \text{Coord}_{1,1} \rightarrow \text{Caus}$ forgets the coordinates and keeps the order (the abstract notion of a causal space goes back to Kronheimer and Penrose [15]); Caus+N is not a formalism in the sense of Section 2 but a formalism together with a sampling model, which is what “number” means in the causal set slogan. The sequence is therefore a translation followed by a statistical step, not a chain in the sense of Definition 2.4. All statements below are about uniform samples of a fixed background; none is a statement about physics.

Proposition 5.1 (Kernel of the order translation). *Every map $(u, v) \mapsto (f(u), g(v))$ with f, g strictly increasing bijections of $(0, 1)$, and the swap $(u, v) \mapsto (v, u)$, preserve the causal order of every configuration. Boosts $(u, v) \mapsto (\lambda u, v/\lambda)$, as maps of $\mathbb{R}^{1,1}$, preserve the order and in addition all proper times; a general pair (f, g) does not.*

Proof. $u_p < u_q \iff f(u_p) < f(u_q)$ for increasing f , and likewise for g ; the swap exchanges the two coordinates in the conjunction defining $\prec$. Under a boost $(\lambda \Delta u)(\Delta v/\lambda) = \Delta u \Delta v$. For $f(u) = u^2$, $g = \text{id}$, the proper time between $(0, 0)$ and (u, v) becomes $u\sqrt{v}$. $\square$

The maps (f, g) with f, g increasing homeomorphisms of $(0, 1)$ are conformal maps of the diamond onto itself that preserve time and space orientation, and together with the swap (parity) they form a group $(\text{Homeo}^+(0, 1))^2 \rtimes \mathbb{Z}_2$ of causal automorphisms of the diamond. The group is infinite-dimensional, which is the well-known reason why Zeeman’s theorem, that the causal automorphisms of Minkowski space are generated by the orthochronous Lorentz group, translations and dilations, requires dimension at least three [37]. In the language of Section 2, $\ker \sigma_1$ contains the conformal orbits, the analogue of the monotone distortions that generate $\ker \tau_2$ in the Euclidean chain, so $\text{Inv}(\sigma_1)$ is contained in conformal geometry; the inclusion is strict at finite n (Proposition 5.2), and equality is the content of the continuum theorems: a bijection between past- and future-distinguishing spacetimes that preserves the chronological

D	n	samples	comparable fraction	expected	dimension estimate
2	400	60	0.4989 ± 0.0185	0.5000	2.00 ± 0.05
3	400	60	0.2298 ± 0.0167	0.2286	3.00 ± 0.09
4	400	60	0.1005 ± 0.0121	0.1000	4.00 ± 0.14

Table 5: E5b. Fraction of comparable pairs (mean $\pm$ sd over samples) in uniform samples of causal diamonds of Minkowski space of dimension D , the expected value, and the Myrheim–Meyer estimate of D .

order in both directions is a smooth conformal isometry [10, 19]. The finite-sample counterparts are tested next.

Proposition 5.2 (Finite non-uniqueness, causal). *Let $n \geq 3$. The kernel class of a configuration under σ_1 is open in $(0, 1)^{2n}$, and it is the union of the conformal orbits, modulo the swap, of the realizers $(\prec_u, \prec_v)$ of its order; in particular it contains configurations that lie in distinct conformal orbits.*

Proof. The order is defined by strict inequalities between coordinates, so it is locally constant and the class is open. Two configurations with the same pair of linear orders $(\prec_u, \prec_v)$ are related by a pair (f, g) of increasing bijections of $(0, 1)$ (interpolate piecewise linearly between the sorted u -values of one and of the other, and likewise for v), and conversely (f, g) preserves both linear orders while the swap exchanges them; so the conformal orbit modulo the swap of a configuration is exactly the set of configurations with the same realizer modulo the swap, and the kernel class is the union of these sets over the realizers of the order. An antichain of n points (all pairs unrelated) has $n!/2$ realizers modulo the swap; for instance three pairwise unrelated points with u -order $(1, 2, 3)$ and v -order $(3, 2, 1)$, and three with u -order $(2, 1, 3)$ and v -order $(3, 1, 2)$, have the same (empty) order and lie in different conformal orbits. $\square$

The u -order is thus a conformal invariant that is not an order invariant, and the causal setting has the same finite/asymptotic structure as the Euclidean one: what the order alone leaves undetermined at finite n is exactly a realizer, and the coordinate reconstruction of E5d below is the choice of a realizer with the help of the counts.

Computational check (E5a). For 200 uniform samples of $n = 300$ points, random pairs (f, g) of increasing bijections preserved the order in all 200 cases while changing the proper times of related pairs by 23% (median over samples of the per-sample median relative change); random boosts preserved the order in all 200 cases and the proper times to within a relative 10^{-9} ; a Euclidean rotation of the (t, x) plane by 5° to 30° , which is not a conformal map, changed the relation of a median 19.6% of the pairs (never fewer than 5.9%).

Order alone: dimension

The fraction of comparable pairs in a uniform sample of a causal diamond is an order invariant with a known expectation in D -dimensional Minkowski space, $\Gamma(D+1)\Gamma(D/2)/(2\Gamma(3D/2))$, twice the Myrheim–Meyer ratio $\langle R \rangle / N^2$ of ordered relations to ordered pairs, which is the basis of the Myrheim–Meyer dimension estimator [21, 22]. Table 5 reports the observed fractions for $n = 400$ points in diamonds of dimension 2, 3 and 4 (60 samples each) next to the expected values, and the dimension estimate obtained by inverting the formula. The three dimensions are separated by many standard deviations. This is the Lorentzian analogue of the count of ordinal patterns in Section 3: a bare order knows the dimension of the space it was sampled from.

n	pairs	$\text{corr}(L, \tau)$	$\text{corr}(L, \tau')$	slope	rel. err. vs τ	rel. err. vs τ'
250	12000	0.967	0.856	1.732	0.143	0.229
500	12000	0.978	0.866	1.739	0.138	0.217
1000	9000	0.988	0.871	1.792	0.104	0.217
2000	6000	0.990	0.909	1.854	0.082	0.166

Table 6: E5c. Longest-chain length L against proper time τ in uniform samples of the 1+1 diamond; τ' is the proper time after a random conformal reparametrisation, under which L is unchanged. Slope: least-squares coefficient through the origin of L on $\sqrt{n}\tau$ (limit 2). Last two columns: median relative error of $\hat{\tau} = L/(2\sqrt{n})$ against τ and against τ' , over pairs with $\tau > \frac{1}{4}$.

Order plus number: proper time

For related points $p \prec q$ let $L(p, q)$ be the number of links of the longest chain from p to q , an order invariant. In the 1+1 diamond a chain is an increasing sequence in both coordinates, so $L(p, q) - 1$ is the length of the longest increasing subsequence of the points inside the rectangle spanned by p and q , whose expected number is $N = (n-2)\tau(p, q)^2$. Uniform points in a rectangle induce a uniformly random permutation, so by the Vershik–Kerov and Logan–Shepp theorem the longest increasing subsequence of N of them has length $2\sqrt{N}(1 + o(1))$ [18, 34]; Brightwell and Gregory [6] and Bollobás and Brightwell [3] give the corresponding statements for sprinklings in Minkowski space of any dimension, and Bollobás and Brightwell [4] the concentration of the chain length. Hence

$$L(p, q) \approx 2\sqrt{n}\tau(p, q),$$

and $\hat{\tau} = L/(2\sqrt{n})$ estimates the proper time from the order and the density alone. Table 6 reports, for n from 250 to 2000 and 39 000 related pairs in total, the correlation between L and τ , the least-squares slope through the origin of L against $\sqrt{n}\tau$, and the median relative error of $\hat{\tau}$. At $n = 2000$ the correlation is 0.990 and the slope 1.85 (it is 1.73 at $n = 250$; the mean length of the longest increasing subsequence is $2\sqrt{N}$ minus a correction of order $N^{1/6}$ with a negative constant [1], which is why all slopes lie below 2); the median relative error of $\hat{\tau}$ over pairs with $\tau > 1/4$ falls from 14% to 8%. The same table shows what “number” contributes: after a random conformal reparametrisation (f, g) the order, and hence L , is unchanged (this is asserted in the code for every pair), but the same $\hat{\tau}$ misses the proper times τ' of the reparametrised configuration by a median of 17% at $n = 2000$ (and their correlation with L drops to 0.91); given E5a this is expected rather than independent evidence, and it is reported as an illustration. The chain length estimates the proper time of the frame in which the sample is uniform: the counting measure selects the conformal factor, and without it the order alone cannot.

Order plus number: coordinates

Under uniform density, $|\downarrow q| \approx (n-1)u_q v_q$ and $|\uparrow q| \approx (n-1)(1-u_q)(1-v_q)$ for the past and future of q , so the order and the count give $s = u_q + v_q$ and $p = u_q v_q$ and hence the unordered pair $\{u_q, v_q\}$ as the roots of $x^2 - sx + p$. Which root is u_q is a sign per point. Both kinds of pairs constrain the relative signs: a comparable pair must satisfy $(u_r - u_q)(v_r - v_q) > 0$ and an unrelated pair $r \parallel q$ must satisfy $(u_r - u_q)(v_r - v_q) < 0$, and whenever exactly one of the two relative orientations satisfies the constraint the pair votes “same” or “opposite” (in practice the votes of comparable pairs change the error by less than one percent); the votes are resolved globally by the leading eigenvector of the vote matrix, and the remaining global swap is the parity of Proposition 5.1, which no order can fix. The result is a reconstruction of every coordinate, not merely of the configuration up to a group, because the boundary of the diamond is part of the sampling hypothesis. Table 7 reports the root-mean-square coordinate error over n from 250 to 2000: the median falls from 0.073 to 0.031, roughly as $n^{-0.39}$ over the range, slower than the $n^{-1/2}$ fluctuation of the counts; the loss is expected from the ill-conditioning

n	samples	median RMSE	max RMSE	median RMSE $\times \sqrt{n}$	order disagreement
250	10	0.0732	0.0971	1.16	7.62%
500	10	0.0494	0.0622	1.10	5.24%
1000	6	0.0415	0.0544	1.31	4.53%
2000	4	0.0312	0.0379	1.39	3.68%

Table 7: E5d. Root-mean-square error of the coordinates (u, v) recovered from the causal order and the past/future counts (after fixing the global swap), and the fraction of pairs whose comparability differs between the true and the reconstructed configuration.

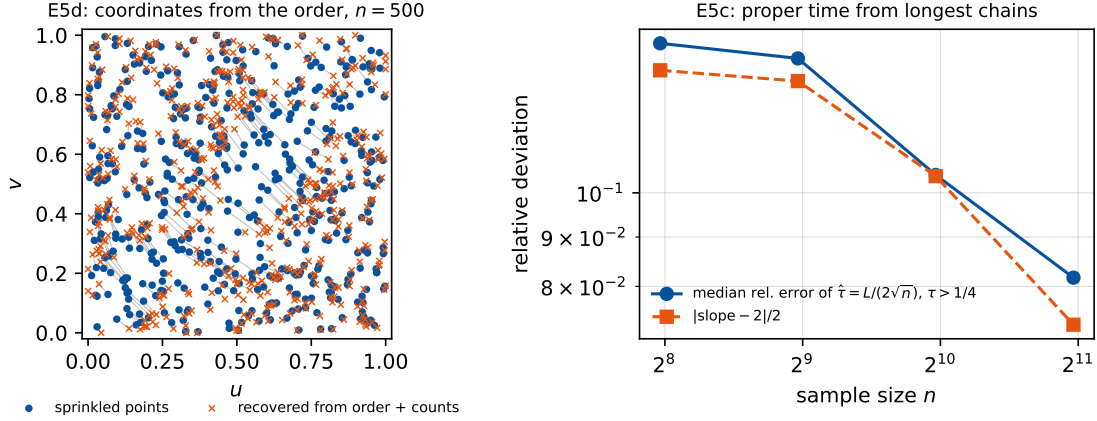


Figure 3: Left (E5d): a uniform sample of the 1+1 diamond and the coordinates recovered from its causal order and counts. Right (E5c): relative error of the chain-length estimate of proper time and relative deviation of the fitted slope from 2, against n .

of the roots near $u = v$, where the discriminant $s^2 - 4p$ vanishes and an $O(n^{-1/2})$ error in the counts becomes an $O(n^{-1/4})$ error in the roots (the rate is not established here). The order recomputed from the reconstructed coordinates disagrees with the true order on 3.68% of the pairs at $n = 2000$. Figure 3 shows one instance.

The dictionary. The two chains line up term by term. Ranking of distances $\leftrightarrow$ causal order; monotone distortion of the metric $\leftrightarrow$ conformal reparametrisation; similarity class $\leftrightarrow$ conformal class; Theorem 3.7 $\leftrightarrow$ Hawking–King–McCarthy and Malament; the density hypothesis that makes non-metric scaling converge $\leftrightarrow$ the counting measure that fixes the conformal factor. The Euclidean chain is the test bed in which every step is a theorem or an exact computation; the Lorentzian chain is where the same steps carry their historical weight. What the present draft does *not* do is state or test anything about curved backgrounds, about dimensions above 1+1 beyond the ordering fraction, or about any physical system.

6 What can be claimed

Table 8 collects the claims of the paper with their status. The list is the deliverable promised by the research line: it separates what is proved from what is measured and from what is only expected.

Limitations. All objects are labelled; quotienting by relabelling turns the kernels into isomorphism problems that are not studied here. The sampling distribution is uniform on a cube or a diamond throughout; the finite- n rates reported (E2, E5c, E5d) are empirical for that

claim	status
Distance matrix determines the configuration up to $E(d)$ (Thm. 3.1)	classical
Betweenness and congruence are empty on generic finite samples, $d \geq 2$ (Prop. 3.2)	proved here
no instance in 4000 exact random configurations; grids rich (E1)	verified
Ordinal fibres have non-empty interior; ordinal $\neq$ similarity at any finite $n \geq 3$ (Prop. 3.5)	proved here
four points realise 120 of 720 orderings on the line, all in the plane (E3)	verified
Ordinal pattern determines the configuration up to similarity as $n \rightarrow \infty$ (Thm. 3.7)	literature
ordinal reconstruction within 2.1×10^{-10} (Procrustes) of X at $n = 256$ in the plane (E2)	verified
Inradius of the ordinal class is $g/4$ (g = minimal gap between sorted distances), in $[g/4, g/2]$ in general (Prop. 3.8)	proved here
median $g \sim n^{-4.2}$, walk radius $\sim n^{-8.0}$ under uniform sampling (E2c)	verified, rates empirical
MST, k -NN graph, RNG, diameter pair are ordinal invariants, for distinct distances (Sec. 4)	proved here
Diameter, affine dimension, Gabriel graph are metric, not ordinal, at finite n (Sec. 4, E4b)	proved here / verified
Kernel of the causal-order translation in 1+1 contains the conformal maps and parity (Prop. 5.1)	proved here
Kernel classes are unions of conformal orbits over realizers; strictly larger than one orbit at finite n (Prop. 5.2)	proved here
Causal order determines topology and conformal structure of the continuum	literature
Ordering fraction estimates the dimension of the diamond (E5b)	literature / verified
Longest chains estimate proper time from order and density, slope $\rightarrow 2$ (E5c)	literature / verified
Coordinates in the diamond recovered from order and counts, error decreasing with n (E5d)	verified; rate not established
Analogous statements for curved backgrounds or physical systems	not addressed

Table 8: Claims and their status. “Verified” means reproduced by the scripts of Appendix A with the fixed seed; it is not a proof.

distribution and are not proved. The ordinal reconstruction uses one solver with random starts; a local minimum would inflate a disparity, never deflate it. The Lorentzian setting is treated only in 1+1 dimensions, where the conformal group is exceptionally large, and only on a flat background; its second step is statistical, not a translation, so its “invariants” are consistent estimators rather than exact invariants.

Next steps for the research line. (a) Prove the n^{-4} law for the minimal gap under uniform sampling (a minimal-spacing statement for weakly dependent distances) and an upper bound on the diameter of the ordinal class in Procrustes disparity, to complement the lower bounds of E2c. (b) Extend E5d to 2+1 dimensions, where the conformal group is finite-dimensional and the count-based reconstruction must be replaced by an embedding of the order. (c) Replace the uniform-sampling hypothesis in Caus+N by a general density and characterise which conformal factors it can and cannot select. None of these is a physical claim.

A Reproducibility

The manuscript directory contains `experiments/formalism_chain.py` (E1–E4), `experiments/ordinal_class.py` (E2c) and `experiments/lorentzian_chain.py` (E5). Both take a `-fast` flag for a reduced run. The full runs used seed 20260930 (E5: 20260930 + 1), Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [35] and scikit-learn 1.9.1; E1–E4 took 1201 seconds, E5 149 seconds and E2c 165 seconds on four cores. The scripts write `results/results.json`, `results/results_lorentz.json` and Markdown tables; `experiments/make_numbers.py` converts them into the macros and table bodies used by this document, so that every number in the text is generated and none is typed.

E1. Coordinates are drawn uniformly from $\{0, \dots, 2^{30} - 1\}^d$; collinearity is tested by the vanishing of all 2×2 minors and strict betweenness by a negative inner product, congruence by equality of squared distances, all in 64-bit integers.

E2. Points uniform in $[0, 1]^d$; rank matrix of the pairwise distances; `sklearn.manifold.MDS` with `metric_mds=False`, precomputed input, random initialisation, four starts, at most 50 000 iterations, stress tolerance 10^{-13} ; Kendall discordance between the input and output rankings via `scipy.stats.kendalltau`; `scipy.spatial.procrustes` for the disparity.

E2c. For each configuration: g and its extremal pairs from the sorted distances; 20 random perturbations with independent uniform coordinates in $(-0.999g/4, 0.999g/4)$; the explicit displacement of Proposition 3.8(ii) with $g/4$ increased by the larger of a relative 10^{-6} and an absolute 10^{-12} , a margin above double-precision rounding of distances of order one; a Gaussian random walk with step size adapted towards an acceptance rate of about 0.3, accepted only when the ranking is unchanged, recording the maximum Procrustes disparity from X .

E3. Patterns are the permutations sorting the six pairwise distances; ties have probability zero under uniform sampling.

E4. $n = 30$, $d = 2$; rigid motions from a QR factorisation of a Gaussian matrix with random reflection; scale factors in $[0.1, 10]$; exponents $p \in [0.3, 3]$; equality of graphs as sets of edges, of vectors to 10^{-9} , of the numerical rank with relative eigenvalue threshold 10^{-6} .

E5. Uniform samples of the unit square in null coordinates (a fixed- n rather than Poisson sprinkling); diamonds in dimension 3 and 4 by rejection from the bounding box; longest chains by patience sorting on the points inside the rectangle spanned by a related pair; coordinate recovery as described in Section 5, with the sign vector taken from the leading eigenvector of the vote matrix.

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